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Can

Abstract

Provided is a can with a shoulder portion decorated in a novel way. A can includes: a trunk portion; a mouth portion smaller in diameter than the trunk portion; and a shoulder portion provided between the trunk portion and the mouth portion, gradually decreasing in diameter from the side of the trunk portion toward the side of the mouth portion, and having a printed layer and a coating film respectively formed by printing and coating. The shoulder portion includes: a straight portion having an inclined cross-sectional shape along a can axis C; a lower curved portion that is a curved portion connecting the straight portion and the trunk portion; an upper curved portion that is a curved portion connecting the straight portion and the mouth portion; and an embossed portion provided only in the straight portion among the straight portion, the lower curved portion, and the upper curved portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Rule 53(b) Continuation of International Application No. PCT/JP2021/021675 filed Jun. 8, 2021, which claims priority based on Japanese Patent Application No. 2020-123320 filed Jul. 17, 2020, the respective disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The disclosure relates to a can with a decorated shoulder portion.

BACKGROUND

(2) There is a known can having a thick-walled shoulder portion decreasing in diameter from a thin-walled cylindrical trunk portion and a mouth portion, which are sealed by double seaming the mouth portion with a can lid or seaming with a metal cap.

(3) The can trunk portion may be decorated by printing or embossing as disclosed in JP 2003-340539 A. On the other hand, as the decoration of a can shoulder portion, there are cases where printing is performed as in JP 2004-168346 A and where an uneven pattern is formed on the shoulder portion as in JP 2004-123231 A, US 2015/0,360,279 A, and Chinese Patent Application Publication No. 103803145.

(4) In recent years, with the reduction in thickness of cans in line with resource saving, in a case where an uneven pattern is formed on the shoulder portion of a thinned can, if a shoulder portion formation die such as that of JP 2004-123231 A (reference sign **60** in FIG. 7) or US 2015/0,360,279 A (reference sign **10** in FIG. 1) is pressed against the shoulder portion of the can, the shoulder portion buckles. In addition, also in a case where the uneven pattern is formed by pressing a formation die such as the groove forming tool of JP 2004-123231 A (reference sign **72** in FIG. 8) only from the outside of the can shoulder portion, the shoulder portion of the thinned can is abnormally deformed.

SUMMARY

(5) The disclosure has been made in view of such circumstances, and the purpose of which is to provide a can that is properly embossed against the shoulder portion of the can.

(6) According to one aspect of the disclosure, there is provided a can including: a cylindrical portion; a small-diameter portion having a diameter smaller than a diameter of the cylindrical portion; and a reduced-diameter portion that is provided between the cylindrical portion and the small-diameter portion, gradually decreases in diameter from a side of the cylindrical portion toward a side of the small-diameter portion, and has a film formed on at least one of an inner surface and an outer surface by printing or coating, in which the reduced-diameter portion includes: an inclined portion having an inclined cross-sectional shape extending through a center axis of the can; a first curved portion that is a curved portion connecting the inclined portion and a can trunk portion; a second curved portion that is a curved portion connecting the inclined portion and the small-diameter portion; and an embossed portion provided only in the inclined portion among the inclined portion, the first curved portion, and the second curved portion.

(7) According to the disclosure, by providing an embossed portion on a shoulder portion of a can, it is possible to decorate the shoulder portion in a new way, and it is possible to suppress defects (such as defective seaming of a can lid) caused by providing the embossed portion on the shoulder portion.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIGS. 1A to 1C are views illustrating a can according to a first embodiment.
- (2) FIG. 2 is an enlarged vertical cross-sectional view of the vicinity of a shoulder portion of the can according to the first embodiment (a view taken along a cutting plane extending through a can axis C).
- (3) FIG. 3A and FIG. 3B are diagrams illustrating a positional relationship between an embossed portion and a printed portion according to the first embodiment.
- (4) FIG. 4A and FIG. 4B are a vertical cross-sectional view of the vicinity of one embossed portion and a table showing measurement value data of dimensions of each portion of the embossed portion, according to the first embodiment.
- (5) FIG. 5A and FIG. 5B are diagrams illustrating an inner roll and an outer roll at the time of embossing according to the first embodiment.
- (6) FIG. 6A and FIG. 6B are enlarged views illustrating an aspect at the time of embossing according to the first embodiment.
- (7) FIGS. 7A to 7E are views illustrating a manufacturing process of the can according to the first embodiment with reference to a schematic view including a partial cross section of the can.
- (8) FIG. 8A and FIG. 8B are enlarged views of the vicinity of a shoulder portion 203 of a can 201 according to a second embodiment.
- (9) FIG. 9 is an enlarged vertical cross-sectional view of the vicinity of the shoulder portion 203 of the can 201 according to the second embodiment (a view taken along a cutting plane extending through a can axis C).
- (10) FIG. 10 is an enlarged vertical cross-sectional view of the vicinity of a shoulder portion 303 of a can 301 of a third embodiment (a view taken along a cutting plane extending through a can axis C).

DESCRIPTION OF EMBODIMENTS

Embodiments

- (11) Hereinafter, embodiments of the disclosure will be described with reference to the drawings and the like.

First Embodiment

- (12) FIGS. 1A to 1C are views illustrating a can 1 according to a first embodiment.
- (13) FIG. 1A is a front view of the entirety of the can 1.
- (14) FIG. 1B is a cross-sectional view taken along line B-B in FIG. 1A, and is a view illustrating a state in which a shoulder portion 3 is viewed from above.
- (15) FIG. 1C is a perspective view of the vicinity of a part of an embossed portion 20 (a portion where “ $\bigcirc \times \Delta$ ” are formed).
- (16) FIG. 2 is an enlarged vertical cross-sectional view of the vicinity of the shoulder portion 3 of the can 1 according to the first embodiment (a view taken along a cutting plane extending through the can axis C).
- (17) FIG. 3A and FIG. 3B are diagrams illustrating a positional relationship between the embossed portion 20 and a printed portion according to the first embodiment.
- (18) FIG. 3A is a view illustrating a part of the embossed portion 20 and a printed portion 6b corresponding thereto when viewed from the normal direction of a straight portion 10.

(19) FIG. 3B is a diagram illustrating another aspect of the printed portion.

(20) In the embodiment, the center axis of the can **1** is appropriately referred to as a can axis C, and the can lid side in the axial direction of the can axis C is appropriately referred to as an upper side, and the bottom side is appropriately referred to as a lower side.

Configuration of Can **1**

(21) The can **1** is formed of a known metal material used for a can, such as steel, tin, aluminum, or an aluminum alloy.

(22) As illustrated in FIGS. **1A** to **1C**, and FIG. **2**, the can **1** includes a trunk portion **2** (cylindrical portion), the shoulder portion **3** (reduced-diameter portion), and a mouth portion **4** (small-diameter portion).

(23) A printed layer **6** (film) is provided on the outer surface of the can **1** by, for example, gravure printing, offset printing, flexographic printing, inkjet printing, or the like, and a coating film **7** (film) is provided on the back surface of the can **1** by coating or the like. Such a film may be formed by thermally bonding a film such as a polyester film to the surface. Further, the printed layer **6** may be provided on such a film after the film is thermally bonded to the can **1**.

(24) As illustrated in FIG. **1B**, a front printed portion **6a** which is printed with a trade name “ $\bigcirc \times \Delta$ ” or the like is provided at the center of the front surface of the outer surface of the trunk portion **2** and at a portion which becomes the front surface when displayed.

(25) The trunk portion **2** has a cylindrical shape. The outer diameter of the trunk portion **2** is usually $\phi 53$ mm, $\phi 66$ mm or the like. The lower end of the trunk portion **2** is provided with a chime portion **2a** whose diameter gradually decreases toward the lower side.

(26) The shoulder portion **3** is connected to the upper side of the trunk portion **2** and is a portion whose diameter gradually decreases toward the upper side (mouth portion **4** side). In other words, the diameter of the shoulder portion **3** decreases from the trunk portion **2** side to the mouth portion **4** side. The maximum outer diameter of the shoulder portion **3** (i.e., the outer diameter of the connecting portion between the shoulder portion **3** and the trunk portion **2**, which is the same as the outer diameter of the trunk portion **2** when the side surface of the trunk portion **2** is straight) may be, for example, from 50 to 70 mm.

(27) The wall thickness of the shoulder portion is preferably as thin as 0.1 to 0.3 mm, and more preferably 0.1 to 0.2 mm. By setting the wall thickness of the shoulder portion in this manner, the material of the can **1** can be reduced.

(28) However, since the shoulder portion **3** is subjected to diameter reduction processing, the wall thickness of the shoulder portion **3** may be thicker than the wall thickness of the trunk portion **2** and the like. In this case, since the shoulder portion **3** has a sufficient strength, it is possible to suppress formation of a fine hole such as a pinhole at the time of processing, and to suppress buckling or the like caused by an external force.

(29) The mouth portion **4** is connected to the upper end side of the shoulder portion **3** and extends upward. A tip of the mouth portion **4** is provided with a flange **5**. A known can lid (not illustrated) is wound around the mouth portion **4**. The inside diameter of the mouth portion **4** is smaller than the diameter of the trunk portion **2**, and can be, for example, from 25 to 60 mm.

(30) The ratio of the shoulder portion maximum outer diameter to the inner diameter of the mouth portion **4** is preferably, for example, from 1.05 to 1.58. By setting such a ratio, the width of the shoulder portion **3** having an installation range **A20** of the embossed portion **20** can be sufficiently wide.

Configuration of Shoulder Portion **3**

(31) As illustrated in FIG. **2**, the shoulder portion **3** includes the straight portion **10** (inclined portion), a lower curved portion **11** (first curved portion), an upper curved portion **12** (second curved portion), and the embossed portion **20**.

(32) In FIG. **2**, both end portions of the lower curved portion **11** (a contact point **11a** between the lower curved portion **11** and the trunk portion **2** and a contact point **11b** between the lower curved

portion **11** and the straight portion **10** of the shoulder portion **3**) and both end portions of the upper curved portion **12** (a contact point **12b** between the upper curved portion **12** and the straight portion **10** and a contact point **12a** between the upper curved portion **12** and the mouth portion **4**) are indicated by black circles. Although the thickness of the shoulder portion **3** gradually increases from the lower curved portion **11** to the straight portion **10** and then to the upper curved portion **12**, each figure shows the thickness equally for simplicity.

(33) The straight portion **10** is a portion having a straight cross-sectional shape.

(34) The straight portion **10** is a portion inclined in a shape of a truncated cone side surface in a middle region of the shoulder portion **3**. The inclination angle of the straight portion **10** with respect to the can axis C is preferably set to from 10 to 50° (more preferably from 25 to 45°). This is because the steeper the inclination of the shoulder portion **3** (closer to vertical), the wider the shoulder portion **3**, and thus the region of the straight portion **10**, which is the installation range A20 of the embossed portion **20**, can be made larger.

(35) The lower curved portion **11** is a curved portion connecting the straight portion **10** and the trunk portion **2**.

(36) The upper curved portion **12** is a curved portion connecting the straight portion **10** and the mouth portion **4**.

(37) In the embodiment, an example in which the lower curved portion **11** is an arc connecting the straight portion **10** and the trunk portion **2** in a tangential line shape in a cross-sectional view is described, but the lower curved portion **11** is not limited thereto. For example, the lower curved portion **11** may be connected at a vertex having a sufficiently large angle (approximately 180°) instead of the tangential line shape, or may be a curved portion having an elliptical shape or the like. The same applies to the upper curved portion **12**.

(38) Among the straight portion **10**, the lower curved portion **11**, and the upper curved portion **12**, the embossed portion **20** is provided only in the straight portion **10**. The shape of the embossed portion **20** is an outward protruding shape as illustrated in FIG. 1C and the like, but may be an inward protruding shape.

(39) The shape of the embossed portion **20** (the shape when the straight portion **10** is viewed from the normal direction) is a shape of the character string of the trade name “◯×Δ” or “ABCSEIKAN”. Therefore, for example, the shape of the embossed portion **20** has a linear portion, a portion having a complicated shape (a curved portion, an intersecting portion, a portion forming a corner, or the like). Further, the cross-sectional shape of the shoulder portion **3**, which is orthogonal to the can axis C and includes the embossed portion **20**, has a shape in which the unevenness changes irregularly.

(40) The shape of the embossed portion is not limited thereto, and various shapes can be employed. For example, the embossed portion may be designed to have a geometric pattern, a symbol, a person, an animal, a plant, a vehicle, an instrument, a landscape, food or drink, packaged food or drink, or the like. As will be described later, the three dimensional shape of the embossed portion may be a truncated cone shape or the like.

(41) On the outer surface of the shoulder portion **3**, the installation range A20 of the embossed portion **20** is separated from the contact point **11a** of the lower curved portion **11** and the trunk portion **2** by 0.8 mm or more in the inner radial direction, and separated from the contact point **12a** of the upper curved portion **12** and the mouth portion **4** by 0.8 mm or more in the outer radial direction.

(42) As described above, the embossed portion **20** is disposed at a position sufficiently away from the lower curved portion **11**. Thus, when an external force is applied to the can **1**, the vicinity of the lower curved portion **11** can be prevented from becoming a starting point of buckling.

(43) Further, the embossed portion **20** is disposed at a position sufficiently away from the upper curved portion **12**. Thus, it is possible to suppress shape defects and the like of the flange **5** caused by forming the embossed portion **20**. As a result, defective seaming of the can lid is suppressed.

- (44) Furthermore, by providing the embossed portion **20** in the straight portion **10**, the embossed portion **20** does not come into contact with other members during handling (during transportation, storage, display in a store, or the like), and the print in the vicinity of the embossed portion **20** does not peel off.
- (45) These functions and effects can be sufficiently expected if the embossed portion **20** is disposed only in the straight portion **10** and is installed without overlapping the lower curved portion **11** and the upper curved portion **12**.
- (46) As illustrated in FIG. 1B, when the can **1** is viewed from the upper side, the installation range **A20** of the embossed portion **20** is a portion corresponding to an arc having a center angle θ of $0^\circ < \theta \leq 260^\circ$ in the shoulder portion **3**. That is, the embossed portion **20** is provided on an arc having a center angle θ of 260° or less (the upper limit value of the center angle θ of the arc is 260°). The reason why the installation range **A20** is limited in this manner is that the installation range **A20** is a range corresponding to a length of an outer peripheral surface of a receiving portion **62** of an inner roll **60**, as described later.
- (47) It is preferable that a center portion of the installation range **A20** faces the front direction of the can **1**. In this case, the can **1** is displayed facing the front, whereby the center of the installation range **A20** faces the front. Thus, when the can **1** is displayed, the embossed portion **20** has good visibility and exhibits high appeal.
- (48) Although not illustrated in the drawings, a plurality of the cans **1** are displayed in the front-rear direction in a store or the like. Since the embossed portion **20** is provided on the inclined straight portion **10**, the embossed portion **20** of the can **1** arranged at the rear side can be visually recognized when observed from the front and obliquely upward. Thus, the can **1** can improve the appeal.
- (49) As indicated by hatching in FIG. 3A, the straight portion **10** includes the printed portion **6b**. The printed portion **6b** is a print of a character string corresponding to the shape of the embossed portion **20**, such as the trade name “ $\bigcirc \times \Delta$ ” or “ABCSEIKAN”, and is disposed surrounding the embossed portion **20**.
- (50) In FIG. 3A, only a portion of the printed portion **6b** corresponding to the trade name “ $\bigcirc \times \Delta$ ” is illustrated, but portions such as “ABCSEIKAN” are also printed in the same manner.
- (51) The outer shape of the printed portion **6b** and the outer shape of the embossed portion **20** are designed to be separated from each other by, for example, 1 mm or more. As a result, even if there is a printing shift at the time of printing or a processing shift at the time of processing the embossed portion **20**, this shift can be made inconspicuous.
- (52) As illustrated in FIG. 3B, the printed portion **6b** may be disposed inside each embossed portion **20**.
- (53) The embossed portion **20** may be disposed on the side of the drinking hole of the can lid in the straight portion **10**. In this case, when a purchaser or the like of the can **1** drinks the beverage, the purchaser or the like naturally sees the embossed portion **20**, thus improving the appeal. On the other hand, the embossed portion **20** may be disposed at a position on the straight portion **10** other than the drinking opening side of the can lid. In this case, the beverage dripping onto the straight portion **10** is less likely to accumulate in the straight portion **10**.
- (54) Detailed Shape of Embossed Portion **20**
- (55) FIG. 4A and FIG. 4B are a vertical cross-sectional view of the vicinity of one embossed portion **20**, and a table showing measurement value data of dimensions of each portion of the embossed portion **20**, according to the first embodiment.
- (56) The table in FIG. 4B shows actual measurement values of dimensions of each portion of the can **1** in the case where the visibility of the embossed portion **20** is acceptable and there is no exposure of the metallic base of the can **1** due to cracking of the printed portion **6b** and the coating film **7**, under the conditions that the thickness of the shoulder portion **3** is from 0.16 to 0.17 mm and protruding amounts **h20** of the embossed portion **20** are 0.10, 0.13, 0.16, 0.19, and 0.30 mm.

- (57) The measurement values in FIG. 4B were obtained by actually making a prototype of the can 1 and evaluating the prototype.
- (58) In FIG. 4A and FIG. 4B, the portions and their dimensions are as follows.
- (59) Protruding amount h_{20} : a protrusion height of the embossed portion 20, which is a length between the outer surface of the straight portion 10 and the top surface 23 of the embossed portion 20 in the normal direction of the straight portion 10.
- (60) Ridge curved surface 21, curvature radius R_{21} : the ridge curved surface 21 is a curved surface connecting the top surface 23 of the embossed portion 20 and the side surface 24 of the embossed portion 20. In the cross-sectional view, the ridge curved surface 21 is an arc connecting the top surface 23 and the side surface 24 in a tangential line shape. In FIG. 4A, a contact point 21a between the ridge curved surface 21 and the top surface 23 and a contact point 21b between the ridge curved surface 21 and the side surface 24 are indicated by black circles. The curvature radius R_{21} is a curvature radius of the ridge curved surface 21.
- (61) Root curved surface 22, curvature radius R_{22} : a curved surface connecting the side surface 24 of the embossed portion 20 and the outer surface of the straight portion 10. In the cross-sectional view, the root curved surface 22 is an arc connecting the straight portion 10 and the side surface 24 in a tangential line shape. In FIG. 4A, a contact point 22a between the root curved surface 22 and the straight portion 10 and a contact point 22b between the root curved surface 22 and the side surface 24 are indicated by black circles. The curvature radius R_{22} is a curvature radius of the root curved surface 22.
- (62) The ridge curved surface 21 may be connected to the top surface 23 and the side surface 24 at a vertex having a sufficiently large angle (approximately 180°) instead of the tangential line shape, or may be a curved portion having an elliptical shape or the like. The same applies to the root curved surface 22.
- (63) R center-to-center distance L_1 (mm): a distance along the straight portion 10 between the contact point 21a and the contact point 22a. In the embodiment, the ridge curved surface 21 and the root curved surface 22 are tangential line shapes. Therefore, the R center-to-center distance L_1 is a distance along the straight portion 10 between the center of the arc of the ridge curved surface 21 and the center of the arc of the root curved surface 22.
- (64) Inclination angle θ_1 ($^\circ$): an inclination angle of the side surface 24 with respect to the straight portion 10.
- (65) From the actual measurement value data in the table in FIG. 4B, when the protruding amount h_{20} of the embossed portion 20 is from 0.10 to 0.30 mm, the dimensions of each portion are in the following ranges. Curvature radius R_{21} (mm) of ridge curved surface 21: from 1.0 to 3.3 Curvature radius R_{22} (mm) of root curved surface 22: from 0.4 to 2.3 R center-to-center distance L_1 (mm): from 1.0 to 1.2 Inclination angle θ_1 ($^\circ$): from 9.4 to 25.4
- (66) It is empirically known from conventional results that the visibility of the embossed portion 20 and the exposure of the metallic base of the can 1 are not affected even if the R center-to-center distance L_1 is increased by about 0.7 mm. In consideration of this, “R center-to-center distance L_1 (mm): from 1.0 to 1.9” can be set.
- (67) Furthermore, based on the actual measurement value data in the table in FIG. 4B, when an approximate curve of the dimensions of each portion was created with the protruding amount h_{20} in the range of from 0.12 to 0.20 mm, the following range was obtained. Curvature radius R_{21} (mm) of ridge curved surface 21: from 1.6 to 2.9 Curvature radius R_{22} (mm) of root curved surface 22: from 0.7 to 1.5 R center-to-center distance L_1 (mm): from 1.0 to 1.2 Inclination angle θ_1 ($^\circ$): from 10.0 to 16.6
- (68) Therefore, within the above range, it can be expected that the visibility of the embossed portion 20 is good and the embossed portion 20 can be processed without exposing the metallic base of the can 1.
- (69) In addition, the outer shape L_2 of the embossed portion 20 can be 1.1 mm or more. This can

be confirmed from the fact that when both a tip of a protruding portion **65** (see FIG. **6A** and FIG. **6B**) of the inner roll **60** and a protruding portion (see an edge portion of a recessed portion **75** in FIG. **6A** and FIG. **6B**) of an outer roll **70** are formed as hemispherical surfaces having a radius of 0.2 mm, the smallest value of the measurement values of the outer shape L2 is 1.1 mm. When the tip of the protruding portion **65** was a hemispherical surface having a radius of 0.3 mm, the smallest value of the actual measurement values of the outer shape L2 was 1.5 mm.

(70) Manufacturing Method for Can **1**

(71) A manufacturing method for the can **1** will be described.

(72) FIG. **5A** and FIG. **5B** are diagrams illustrating the inner roll **60** and the outer roll **70** at the time of embossing according to the first embodiment.

(73) FIG. **5A** is a cross-sectional view taken along line A-A in FIG. **5B**, and is a transverse cross-sectional view of the inner roll **60** and the outer roll **70**.

(74) FIG. **5B** is a view of the inner roll **60** and the outer roll **70** viewed from the outside in the radial direction.

(75) In FIG. **5A** and FIG. **5B**, only some of a plurality of the protruding portions **65** and recessed portions **75** are illustrated, and the other protruding portions **65** and recessed portions **75** are not illustrated.

(76) FIG. **6A** and FIG. **6B** are enlarged views illustrating an aspect at the time of embossing according to the first embodiment.

(77) FIG. **6A** is an enlarged view of a portion indicated by an arrow **6A** in FIG. **5A**.

(78) FIG. **6B** is an enlarged view of a portion indicated by an arrow **6B** in FIG. **5B**.

(79) FIGS. **7A** to **7B** are views illustrating a manufacturing process of the can **1** according to the first embodiment with reference to a schematic view including a partial cross section of the can **1**.

(80) In the manufacturing method of the can **1**, as a pre-process, a bottomed cylindrical intermediate formed body having the trunk portion **2** is manufactured by known drawing and ironing or the like, and the inner and outer surfaces of the intermediate formed body are subjected to printing, coating or the like as necessary. In the embodiment, the intermediate formed body is also referred to as the can **1**.

(81) After that, the intermediate formed body of the can **1** is subjected to known processing such as die necking including a plurality of steps, roll necking (spin flow necking), or a combination of die necking and roll necking including a plurality of steps, thereby forming the shoulder portion **3**.

(82) Then, the mouth portion **4** having the flange **5** at an opening end is formed on the intermediate formed body of the can **1** by a known die flanger, spin flanger or the like.

(83) Thus, an intermediate formed body of the can **1** as illustrated in FIGS. **7A** to **7D** and the like is manufactured.

(84) Next, as illustrated in FIG. **5A** and FIG. **5B**, FIG. **6A** and FIG. **6B**, and FIGS. **7A** to **7E**, a manufacturing apparatus **50** (can manufacturing apparatus) is used to form the embossed portion **20** in the shoulder portion **3**.

(85) Manufacturing Apparatus **50**

(86) The manufacturing apparatus **50** includes a mounting table **51**, the inner roll **60**, and the outer roll **70**.

(87) As illustrated in FIGS. **7A** to **7E**, the mounting table **51** is driven by a driving apparatus (not illustrated) including a motor and the like.

(88) The mounting table **51** is rotationally driven together with the can **1** in a state where the can **1** is mounted thereon. The mounting table **51** is driven to move the can **1** to a pre-processing position and a retracted position retracted from the pre-processing position in a state where the can **1** is mounted on the mounting table **51**. The rotation axis of the mounting table **51**, the rotation axis of the inner roll **60**, and the rotation axis of the outer roll **70** are parallel to each other.

(89) As illustrated in FIG. **5A** and FIG. **5B**, and FIG. **6A** and FIG. **6B**, the inner roll **60** and outer roll **70** form part of a can making tool set.

(90) The inner roll **60** includes a shaft **61** and the receiving portion **62**.

(91) The shaft **61** is rotationally driven by a driving apparatus (not illustrated).

(92) Since the receiving portion **62** is fixed to a lower end of the shaft **61**, the receiving portion **62** is rotationally driven integrally with the shaft **61** (see the rotational direction $\theta 62$). The receiving portion **62** is a truncated cone member having a diameter increasing from the upper side to the lower side. A truncated cone side surface **63** of the receiving portion **62** has an inclination angle along the straight portion **10** of the can **1**.

(93) The maximum outer diameter of the receiving portion **62** is smaller than the inner diameter of the mouth portion **4** of the can **1**. Therefore, the inner roll **60** can be inserted into and removed from the mouth portion **4** of the can **1** at the time of processing the embossed portion **20**.

(94) The truncated cone side surface **63** is provided with the plurality of protruding portions **65** for forming a plurality of the embossed portions **20**.

(95) The protruding portions **65** are provided protruding toward the outside in the radial direction from the truncated cone side surface **63**. The entire periphery of the edge portion of the top surface of the protruding portion **65** is a curved surface **65a**.

(96) Similarly to the inner roll **60**, the outer roll **70** includes a shaft **71** and a receiving portion **72**, and is rotationally driven by a driving apparatus (not illustrated) (see a rotation direction $\theta 72$).

(97) The receiving portion **72** is a truncated cone member having a diameter decreasing from the upper side to the lower side. The receiving portion **72** is larger than the receiving portion **62** of the inner roll **60**. An inclination angle of a truncated cone side surface **73** of the receiving portion **72** is equal to the inclination angle of the truncated cone side surface **63** of the receiving portion **62**.

(98) The truncated cone side surface **73** is provided with the plurality of recessed portions **75** corresponding to the protruding portions **65** of the receiving portion **62**. That is, each protruding portion **65** of the receiving portion **62** and each recessed portion **75** of the receiving portion **72** corresponding thereto are disposed at such positions where they mesh with each other in a case where the receiving portions **62**, **72** rotate at the same peripheral speed.

(99) An edge portion **75a** of an opening of the recessed portion **75** is a curved surface.

(100) In a case where the embossed portion **20** has an inwardly protruding shape, a plurality of recessed portions may be formed in the receiving portion **62** of the inner roll **60** and a plurality of protruding portions may be formed in the receiving portion **72** of the outer roll **70**.

(101) According to the above-described configuration, the shape of the receiving portions **62**, **72** is an umbrella shape having the truncated cone side surfaces **63**, **73** conforming to the shape of the straight portion **10** of the shoulder portion **3**. Accordingly, in a state in which the straight portion **10** is sandwiched by the receiving portions **62**, **72**, these three members are rotationally driven at the same peripheral speed, thereby the receiving portions **62**, **72** rotate in a state in which the straight portion **10** is sandwiched in a bevel-gear-like manner.

(102) As illustrated in FIG. **6A** and FIG. **6B**, and the like, the protruding portion **65** of the receiving portion **62** presses the straight portion **10** of the can **1** from the inside, and the truncated cone side surface **73** of the receiving portion **72** of the outer roll **70** receives the straight portion **10** from the outside.

(103) Therefore, in a region where the protruding portion **65** and the recessed portion **75** are formed, the top surface of the protruding portion **65** pushes the straight portion **10** into the recessed portion **75** while stretching the straight portion **10**. As a result, the embossed portion **20** is formed by drawing.

(104) In this case, the ridge curved surface **21** of the embossed portion **20** has a shape following the curved surface **65a** of the protruding portion **65**. Further, the straight portion **10** is pushed into the recessed portion **75** while abutting against the edge portion **75a** of the recessed portion **75**. Therefore, a portion of the straight portion **10** in the vicinity of the edge portion **75a** is pushed into the recessed portion **75** while being curved. As a result, the root curved surface **22** having a smaller curvature radius than the ridge curved surface **21** is formed.

(105) Note that the edge portion **75a** of the recessed portion **75** may have a rectangular shape.

(106) As described above, the embossed portion **20** is processed into a protruding shape by stretching a part of the straight portion **10**. Accordingly, the printed layer **6** and the coating film **7** (see FIG. 2) provided on the shoulder portion **3** are also stretched together. In this case, when cracks or the like occur in the printed layer **6** and the coating film **7**, the metallic base of the can **1** is exposed. In the embodiment, it is possible to suppress such exposure of the metallic base by setting the dimensions of the respective portions of the embossed portion **20** in the above-described range.

(107) The inclination angle of the straight portion **10** of the can **1** of the embodiment is set to be from 10° to 50°. Therefore, the truncated cone side surface **63** of the receiving portion **62** of the inner roll **60** can secure a sufficient area for forming the plurality of embossed portions **20**. In addition, the inner roll **60** can be inserted into and removed from the mouth portion **4** even if the shaft **61** has a sufficient thickness or wall thickness for strength. Further, when sandwiching the straight portion **10** from the radial direction, the receiving portions **62**, **72** can apply a sufficient pressure to the straight portion **10** from the normal direction of the straight portion **10**. Thus, the processability of the embossed portion **20** can be improved.

(108) Manufacturing Process

(109) A series of manufacturing processes for manufacturing the can **1** will be described.

(110) Can Placing Step: FIG. 7A

(111) The can **1** is mounted on the mounting table **51** by a conveying apparatus (not illustrated).

(112) Inner Roll **60** Insertion Step: FIG. 7B

(113) Next, the can **1** is moved to the processing position by moving the mounting table **51**.

(114) Thus, the inner roll **60** is inserted into the can **1** from the mouth portion **4**.

(115) Sandwiching Step of Sandwiching Shoulder Portion **3**: FIG. 7C

(116) By relatively approaching the inner roll **60** and the outer roll **70** to the shoulder portion **3** of the can **1**, the shoulder portion **3** is sandwiched by the receiving part **62**, **72**. That is, the protruding portion **65** of the receiving portion **62** presses the straight portion **10** of the can **1** from the inside, and the receiving portion **72** of the outer roll **70** receives the straight portion **10** from the outside.

(117) Although detailed description is omitted, the manufacturing apparatus **50** detects the marking printed on the outer surface of the can **1** with an optical sensor. Then, based on the detection result, the manufacturing apparatus **50** drives the mounting table **51** to rotate and position the can **1** so that the processing position of the embossed portion **20** becomes a position corresponding to the printed portion **6b**.

(118) Rotation Step: FIG. 7C

(119) The inner roll **60** and the outer roll **70** are rotated in a state where the shoulder portion **3** is sandwiched by the receiving portions **62**, **72** in the sandwiching step, and the mounting table **51** and the can **1** are integrally rotated.

(120) At this time, the shoulder portion **3** is reliably supported by the receiving portion **72** of the outer roll **70**. Therefore, the shoulder portion **3** of the can **1** is less prone to be abnormally deformed, damaged or the like even if it is thin.

(121) When the inner roll **60** rotates by one rotation, the inner roll **60**, the outer roll **70**, and the mounting table **51** stop rotating. As a result, the embossed portion **20** corresponding to the circumferential surface of one rotation of the receiving portion **62** of the inner roll **60** is formed on the shoulder portion **3**.

(122) Roller Retracting Step: FIG. 7D

(123) The inner roll **60** and the outer roll **70** are separated from the shoulder portion **3** in the radial direction. Thus, in the height direction of the can **1**, the inner roll **60** and the outer roll **70** are retracted to a position where they do not interfere with the can **1**.

(124) Retracting Step: FIG. 7E

(125) By moving the mounting table **51**, the can **1** is relatively separated from the processing

position. Thus, the can **1** is retracted from the processing position.

(126) The inner roll **60** and the outer roll **70** move relative to the can **1** by moving toward the mouth portion **4** side in the height direction. As a result, the inner roll **60** moves from the mouth portion **4** to the outside of the can **1**.

(127) As described above, in the can manufacturing method according to the present embodiment, the shoulder portion **3** is formed into a three dimensional shape while the receiving portion **62** of the inner roll **60** receives the shoulder portion **3** from the inner side, so that damage to the shoulder portion **3** can be suppressed.

(128) As described above, in the can **1** of the present embodiment, the embossed portion **20** can be provided in the shoulder portion **3** while suppressing defects and the like. Thus, the can **1** having a new and unconventional form may be provided.

Second Embodiment

(129) Next, a second embodiment of the disclosure will be described.

(130) In the following description and drawings, the same names, the same reference signs, or the same reference signs at the end (last two digits) are appropriately given to portions that perform the same functions as those according to the first embodiment described above, and redundant description is appropriately omitted.

(131) FIG. **8A** and FIG. **8B** are enlarged views of the vicinity of a shoulder portion **203** of a can **201** according to the second embodiment.

(132) FIG. **8A** is a view illustrating the vicinity of the shoulder portion **203** as viewed from the front (in a direction perpendicular to the can axis C).

(133) FIG. **8B** is a perspective view of the vicinity of the shoulder portion **203**.

(134) FIG. **9** is an enlarged vertical cross-sectional view of the vicinity of the shoulder portion **203** of the can **201** according to the second embodiment (a view taken along a cutting plane extending through a can axis C).

(135) In FIG. **8A** and FIG. **8B**, and FIG. **9**, the step height (height difference) or the like of a stepped portion **210** is illustrated in an exaggerated manner as appropriate.

(136) FIG. **9** illustrates the surface (surface on the outer peripheral side) of the can **201**, and an illustration of the thickness portion of the can **201** is omitted.

(137) In FIG. **9**, a thick line indicates the shape of an embossed portion **220**, a thin line indicates the shape of the stepped portion **210**, and a broken line **210a** indicates the shape of the stepped portion **210** before embossing.

(138) Although detailed description is omitted, the can **201** of the embodiment includes the front printed portion **6b** (see FIG. **3A** and FIG. **3B**, and the like), the coating film **7** (see FIG. **2** and the like), and the like similar to those of the first embodiment.

(139) As illustrated in FIG. **8A** and FIG. **8B**, and FIG. **9**, in the can **201** of the present embodiment, the shoulder portion **203** is provided with the stepped portion **210** (inclined portion) with a series of uneven shapes. That is, the shape of the shoulder portion **203** in a vertical cross-sectional view (a view taken along a cutting plane extending through the can axis C) is a straight shape according to the first embodiment, whereas it is stepped with a series of uneven shapes in the present embodiment.

(140) The stepped portion **210** may have a series of constant uneven shapes or a series of different uneven shapes. In the present embodiment, the former example will be described.

(141) The stepped portion **210** has a shape in which the size in the radial direction (that is, the height of the step) varies depending on the position in the direction along the shoulder portion **203**. Further, the cross-sectional shape of the step is constant or substantially constant in any cross section as long as the cross section is a longitudinal cross section (a cutting plane extending through the can axis C).

(142) In addition, the shape of the can **201** is the same as that of the first embodiment except for the stepped portion **210**, and the can **201** has the same function and effect as those of the first

embodiment.

(143) The shoulder portion **203**, which is a reduced-diameter portion, is processed by a method similar to that for the reduced-diameter portion of a frame can with a multi-stage neck-in portion (for example, JP-A-8-224625). Since the step of the stepped portion **210** is formed at a joint portion of the shoulder portion **203** by this processing, the height (height difference) thereof is very small. Therefore, the stepped portion **210** has a shape that can be regarded as a substantially truncated cone side surface shape.

(144) The shape of the embossed portion **220** of the present embodiment corresponding to that of the first embodiment can be defined as follows.

(145) An alternate long and short dash line in FIG. **9** is a straight line connecting upper and lower contact points **222a** of the root curved surface **222** and the stepped portion **210**, and is a reference line BL for defining dimensions of each portion as described later.

(146) Protruding amount **h220**: corresponds to the protruding amount **h22** (see FIG. **4A** and FIG. **4B**). It is the maximum height of the embossed portion **220** from the reference line BL. In other words, it is the length of the longest line among perpendicular lines from the embossed portion **220** to the reference line BL.

(147) Ridge curved surface **221**, curvature radius **R21**: the ridge curved surface **221** corresponds to the ridge curved surface **21** (see FIG. **4A** and FIG. **4B**). The ridge curved surface **221** is a curved surface connecting a top surface **223** of the embossed portion **220** and a side surface **224** of the embossed portion **220**. The curvature radius **R21** corresponds to the curvature radius of the ridge curved surface **221**.

(148) Root curved surface **222**, curvature radius **R22**: the root curved surface **222** corresponds to the root curved surface **22** (see FIG. **4A** and FIG. **4B**). The root curved surface **222** is a curved surface connecting the side surface **224** of the embossed portion **220** and the surface of the stepped portion **210**. The curvature radius **R22** corresponds to the curvature radius of the root curved surface **222**.

(149) Inclination angle $\theta 201$ ($^{\circ}$): an inclination angle of the side surface **224** with respect to the reference line BL.

(150) R center-to-center distance **L201**: a distance along the reference line BL between a contact point **221a** and a contact point **222a**. As in the first embodiment, the ridge curved surface **221** and the ridge curved surface **221** have a tangential line shape. Therefore, the R center-to-center distance **L1** is a distance along the reference line BL between the center of the arc of the ridge curved surface **221** and the center of the arc of the root curved surface **222**.

(151) Outer shape **L202** of embossed portion **220**: an outer shape of the embossed portion **220**, and the length between the upper and lower contact points **222a**, that is, the length of a straight line connecting the upper and lower contact points **222a**.

(152) As described above, even if the stepped portion **210** does not have a strictly straight shape, the level difference is sufficiently small. Therefore, in a case where the protruding amount **h220** of the embossed portion **220** is from 0.13 to 0.19 mm, the stepped portion **210** can be expected to have the same function and effect as those of the first embodiment with respect to suppression of cracking or the like of the printed portion **6b** and the coating film **7** (see FIG. **1A** and FIG. **1B**, FIG. **2**, and FIG. **3A** and FIG. **3B**) by setting the dimensions of the respective portions to the same dimensions as those of the first embodiment (see FIG. **4B**).

(153) In the above description and FIG. **9**, the upper shape of the embossed portion **220** has been described, but the shape of each portion of the lower shape of the embossed portion **220** also needs to satisfy the above dimensional conditions. In addition, in a case where it is clear which of the upper shape and the lower shape of the embossed portion **220** causes the crack or the like of the printed portion **6b** or the coating film **7** (for example, in a case where it is clear which of the upper shape and the lower shape is larger in the elongation of the raw material accompanying the embossing), only one of the dimensions may be managed.

(154) As described above, in the can **201** of the present embodiment, even if the stepped portion **210** is formed in the shoulder portion **203**, since the step difference is minute, it can be expected that the same operation and effect as those of the first embodiment can be obtained.

Third Embodiment

(155) Next, a third embodiment of the disclosure will be described.

(156) FIG. **10** is an enlarged vertical cross-sectional view of the vicinity of a shoulder portion **303** of a can **301** according to the third embodiment (a view taken along a cutting plane extending through the can axis C).

(157) As illustrated in FIG. **10**, in the can **301** of the present embodiment, a curved portion **310** (inclined portion) is provided in the shoulder portion **303**.

(158) In FIG. **10**, a thick line indicates the shape of an embossed portion **320**, a thin line indicates the shape of the curved portion **310**, and a broken line **310a** indicates the shape of the curved portion **310** before embossing.

(159) The curved portion **310** is gently curved to have a protruding shape toward the inside of the can. The curved portion **310** may be gently curved to have a protruding shape toward the outside of the can.

(160) That is, in the second embodiment, the stepped portion **210** is formed at the shoulder portion, whereas in the present embodiment, the curved portion **310** is formed at the shoulder portion **303**.

(161) The reference line BL, a contact point **321a**, a contact point **322a**, a protruding amount **h320**, a ridge curved surface **321**, the curvature radius **R21**, a root curved surface **322**, the curvature radius **R22**, an inclination angle $\theta 301$, an R center-to-center distance **L301**, an outer shape **L302** of the embossed portion **320**, and the like can be defined in the same manner as in the second embodiment.

(162) Also in the present embodiment, by making the curvature of the shape of the curved portion **310** sufficiently large, it can be expected that the same operation and effect as those of the first embodiment can be obtained.

(163) Although the embodiments of the disclosure have been described above, the disclosure is not limited to the above-described embodiments. For example, various modifications and changes such as modifications to be described later are possible, and they are also within the technical scope of the disclosure. Further, the effects described in the embodiments are merely the most preferable effects generated from the disclosure, and the effects according to the disclosure are not limited to those described in the embodiments. It should be noted that the configurations of the above-described embodiment and modifications to be described later can be used in combination as appropriate, but detailed description thereof will be omitted.

(164) Modifications

(165) (1) In the embodiment, an example in which the shape of the embossed portion is a shape of a character string or the like having a linear portion, a portion having a complicated shape (a curved portion, an intersecting portion, a portion forming a corner, or the like), or the like has been described, but the configuration is not limited thereto. At least some of the embossed portions may be in the form of small truncated cones, cones, or the like. In this case, one embossed portion is observed as one dot. Thus, the geometric shape of a character or the like can be expressed as a plurality of the dot patterns by the plurality of embossed portions.

(166) Also in this case, the outer shape of the printed portion corresponding to each embossed portion and the outer shape of each embossed portion may be separated from each other by, for example, 1 mm or more. Furthermore, the printed portion (printed portion of characters or the like) corresponding to the dot pattern shape expressed by the plurality of embossed portions and the outer shape of the embossed portion may be separated from each other by, for example, 1 mm or more.

(167) Also in this case, the embossed portion may be disposed in the installation range described in the embodiment, and may have the shape described in the embodiment. For example, by setting the

curvature radius R21 of the ridge curved surface 21, the curvature radius R22 of the root curved surface 22, the R center-to-center distance L1, and the inclination angle $\theta 1$ of the embossed portion as described in the embodiment, the embossed portion can be expected to be visible, and exposure of the metallic base can be suppressed. (2) In the embodiment, an example is illustrated in which the installation range of the embossed portion is an arc-shaped portion (see FIG. 1B) of the shoulder portion having a center angle θ of $0^\circ < \theta \leq 260^\circ$, but the configuration is not limited thereto. In the case of a continuous pattern in which the shape of the embossed portion regularly changes (for example, a pattern in which recessed and protruding shapes regularly continue such as in a so-called diamond cut), the installation range of the embossed portion may be a portion corresponding to an arc in which the center angle θ is 360° or less ($0^\circ < \theta \leq 360^\circ$).

(168) That is, in a case where the shape of the embossed portion changes regularly, the cross-sectional shape of the shoulder portion that is orthogonal to the can axis and includes the embossed portion has a shape in which the unevenness changes regularly. Thus, the inner roll and the outer roll can be provided with an uneven shape such that they continuously mesh with each other even when the inner roll rotates one turn or more. Thus, when the inner roll is rotated one turn or more at the time of embossing, the embossed portion can be formed in the arc-shaped portion having the center angle of 360° or less. (3) The can may be a threaded can (such as a bottle can) in which a jaw portion, a thread portion, a curl portion, or the like is formed on the mouth portion of the can having a reduced diameter by providing a thread portion forming step after embossing. Further, the can may be a three-piece can in which the bottom portion, the trunk portion, and the lid portion are different members. (4) In the embodiment, an example is illustrated in which the embossed portion is formed in the straight portion of the shoulder portion, but the configuration is not limited thereto. For example, in a can provided with a constriction having a straight portion in the vicinity of the center of the trunk portion or the like, the embossed portion may be provided in the constriction. Further, in a can provided with a chime portion having a straight portion, the embossed portion may be provided in the chime portion. (5) In the embodiment, an example of the printed layer formed on the outer surface of the can surface of the can and the coating film formed on the inner surface of the can is illustrated, but the configuration is not limited thereto. At least one of the printed layer and the coating film may be provided on at least one of the outer surface and the inner surface of the can.

(169) While preferred embodiments of the disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure. The scope of the disclosure, therefore, is to be determined solely by the following claims.

Claims

1. A can comprising: a cylindrical portion; a small-diameter portion having a diameter smaller than a diameter of the cylindrical portion; and a reduced-diameter portion that is provided between the cylindrical portion and the small-diameter portion, gradually decreases in diameter from a side of the cylindrical portion toward a side of the small-diameter portion, and has a film formed on at least one of an inner surface and an outer surface by printing or coating, wherein the reduced-diameter portion includes an inclined portion having an inclined cross-sectional shape extending through a center axis of the can; a first curved portion that is a curved portion connecting the inclined portion and a can trunk portion; a second curved portion that is a curved portion connecting the inclined portion and the small-diameter portion; and an embossed portion provided only in the inclined portion among the inclined portion, the first curved portion, and the second curved portion, wherein the embossed portion includes a top surface; a side surface; a ridge curved surface which is a curved surface forming a ridge line between the top surface and the side surface; and a root curved surface which is a curved surface connecting the side surface and an outer

- surface of the inclined portion, and wherein a protrusion height of the embossed portion, which is a length between an outer surface of the inclined portion and the top surface of the embossed portion in the normal direction of the inclined portion, is 0.3 mm or less.
2. The can according to claim 1, wherein the embossed portion is formed at a position away from an outer surface of a connecting portion between the first curved portion and the cylindrical portion by 0.8 mm or more in an inner radial direction, and is formed at a portion away from an outer surface of a connecting portion between the second curved portion and the small-diameter portion by 0.8 mm or more in an outer diameter direction.
 3. The can according to claim 1, wherein a cross-sectional shape orthogonal to the center axis of the can and including the embossed portion of the reduced-diameter portion has an irregularly varying shape, and an installation range of the embossed portion in a circumferential direction in the reduced-diameter portion is a range corresponding to an arc having a center angle θ of $0^\circ < \theta \leq 260^\circ$.
 4. The can according to claim 1, wherein a cross-sectional shape orthogonal to the center axis of the can and including the embossed portion of the reduced-diameter portion has a regularly varying shape, and an installation range of the embossed portion in a circumferential direction in the reduced-diameter portion is a range corresponding to an arc having a center angle θ of 360° or less.
 5. The can according to claim 1, wherein a maximum outer shape measurement of the embossed portion is greater than or equal to 1.1 mm.
 6. The can according to claim 1, wherein the protrusion height of the embossed portion is from 0.10 to 0.30 mm, and an inclination angle of the side surface of the embossed portion with respect to the inclined portion is from 9.4° to 25.4° .
 7. The can according to claim 6, wherein the ridge curved surface has a curvature radius of from 1.0 to 3.3 mm.
 8. The can according to claim 6, wherein the root curved surface has a curvature radius of from 0.4 to 2.3 mm.
 9. The can according to claim 6, wherein in a cross-sectional shape extending through a can axis, a length along the inclined portion between an arc center of the ridge curved surface and an arc center of the root curved surface connecting the side surface of the embossed portion and the inclined portion is from 1.0 to 1.9 mm.
 10. The can according to claim 1, wherein a center of the installation range of the embossed portion and a front center of the can are aligned.
 11. The can according to claim 1, wherein the reduced-diameter portion includes a printed portion having a pattern corresponding to a shape of the embossed portion, and an outer shape of the pattern is disposed surrounding an outer shape of the embossed portion, or is disposed inward from the outer shape of the embossed portion.
 12. The can according to claim 1, wherein the inclined portion is a straight portion having a straight cross-sectional shape.
 13. The can according to claim 1, wherein the inclined portion is a stepped portion having a shape in which unevenness is repeated in a cross-sectional shape.
 14. The can according to claim 1, wherein the inclined portion is a curved portion having a cross-sectional shape curved inwardly or outwardly from the can.
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